



POWER BI MAJOR CAPSTONE PROJECT
END TO END
POWER BI PROJECT GUIDE

PROJECT TITLE	Student Social Media & Mental Health Analytics
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PROJECT CATEGORY	Education Analytics Mental Health Analytics Behaviour Analytics
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INSTITUTE NAME	Infinity Code Nexus
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TRAINER / GUIDE	Anamaina Anand Kumar
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CREATED BY	Macha Ramyasree
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TOOLS USED	Power BI Desktop Power Query, DAX & Power BI Service
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SUBMISSION DATE	[14/06/2026]
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Prepared as part of the Power BI Masterclass Program to demonstrate skills in
Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX,
Storytelling, and Business Intelligence Reporting.

PROJECT OVERVIEW

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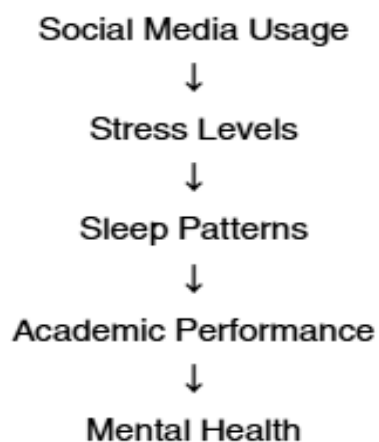
PROJECT OBJECTIVE

Develop an interactive dashboard to analyze the impact of social media usage on student mental health.

Participants are expected to:

- ✓ Analyze student behavior
- ✓ Measure social media usage
- ✓ Evaluate stress levels
- ✓ Understand mental health patterns
- ✓ Generate actionable insights
- ✓ Present recommendations

Convert raw student data into meaningful insights that help understand the relationship between.



PROJECT GOAL

To analyze the impact of social media usage on students' mental health, stress levels, sleep patterns, and academic performance using interactive Power BI dashboards.

BUSINESS PROBLEM

Educational institutions and student counselors want to understand:

The objective is to identify:

1. How social media affects mental well-being.
2. How screen time impacts academic performance.
3. How stress levels vary across different student groups.
4. How sleep patterns influence mental health.
5. Which social media platforms have the highest usage.

BUSINESS QUESTIONS

Participants must answer:

1. What is the average mental health score?
2. What is the average social media usage per day?
3. Which platform is most used?
4. Which student groups report higher stress levels?
5. How does sleep affect mental health?
6. How does study time compare with social media usage?
7. Which countries show higher mental health scores?

DATASET INFORMATION

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DATASET NAME : Student Social Media and Mental Health Impact

DATASET DETAILS : Approximate Records 5000+

Columns : 13

EXPECTED ANALYSIS AREAS

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Executive Analytics

Social Media Analytics

Mental Health Analytics

Life Style Analytics

Country Analytics

POWER QUERY TRANSFORMATIONS

DATA CLEANING ACTIVITIES

- Missing Value Handling
- Duplicate Data Removal
- Data Type Correction
- Data Standardization
- Data Validation & Quality Check

DATA PREPARATION ACTIVITIES

- ✓ Renamed Columns
- ✓ Created Custom Columns
- ✓ Replaced Values
- ✓ Removed Unnecessary Columns

CUSTOM COLUMNS CREATED

Stress Level

Screenshot:

Final Project power bi

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent * Enter Data * Data source settings * Data Sources * Parameters * Export query results * Refresh Preview * Properties * Advanced Editor * Choose Columns * Remove Columns * Keep Rows * Remove Rows * Sort * Split Column * Group By * Data Type: Whole Number * Use First Row as Headers * Replace Values * Merge Queries * Append Queries * Combine Files * Combine

Queries [1]

Student Social Media An...

Table.TransformColumnTypes(*Promoted headers*,{{"Age", Int64.Type}, {"Gender", type text}, {"Country", type text}, {"Academic_Level", type text}, {"Most_Used_Platform", type text}, {"Purpose_Of_Use", type text}, {"Avg_Daily_Usage", type decimal(1,2)}})

	Age	Gender	Country	Academic_Level	Most_Used_Platform	Purpose_Of_Use	Avg_Daily_Usage
1	21 Male	Other	Undergraduate	Facebook	Networking		
2	23 Female	Other	Graduate	Linkedin	Education		
3	22 Male	Canada	Undergraduate	Instagram	Entertainment		
4	18 Male	Other	High School	Snapchat	Entertainment		
5	24 Female	Other	Graduate	Facebook	Networking		
6	19 Female	Other	Undergraduate	Twitter	News		
7	22 Female	Other	Undergraduate	Instagram	Entertainment		
8	21 Female	Other	Undergraduate	Facebook	Networking		
9	24 Male	USA	Graduate	YouTube	Education		
10	19 Male	Other	Undergraduate	TikTok	Entertainment		
11	18 Male	Other	High School	Linkedin	Education		
12	18 Male	Other	High School	Snapchat	Entertainment		
13	20 Male	Other	Undergraduate	Facebook	Networking		
14	23 Female	Other	Graduate	Linkedin	Education		
15	24 Male	India	Graduate	Instagram	Entertainment		
16	18 Female	Other	High School	Twitter	News		
17	22 Female	USA	Undergraduate	YouTube	Education		
18	21 Male	Australia	Undergraduate	YouTube	Education		
19	21 Male	Canada	Undergraduate	Facebook	Networking		
20	20 Male	Other	Undergraduate	Instagram	Entertainment		
21	23 Female	Other	Graduate	Facebook	Networking		
22	19 Female	Other	Undergraduate	Snapchat	Entertainment		
23	24 Male	Other	Graduate	Facebook	Networking		
24	23 Male	India	Graduate	TikTok	Entertainment		
25	23 Female	Other	Graduate	Linkedin	Education		

Query Settings

PROPERTIES

Name

Student Social Media And Mental Health

APPLIED STEPS

Source

Promoted headers

Changed column type

DATA MODELING

Star Schema

Purpose:

FACT TABLE

Basic Details of Student Social Media & Mental Health Data

DIMENSION TABLES

Dim_Country

Dim_Gender

Dim_Academic Level

Dim_Stress Level

Dim_Most Used platform

RELATIONSHIPS

One-to-Many Relationships

FILTER DIRECTION

Single Direction

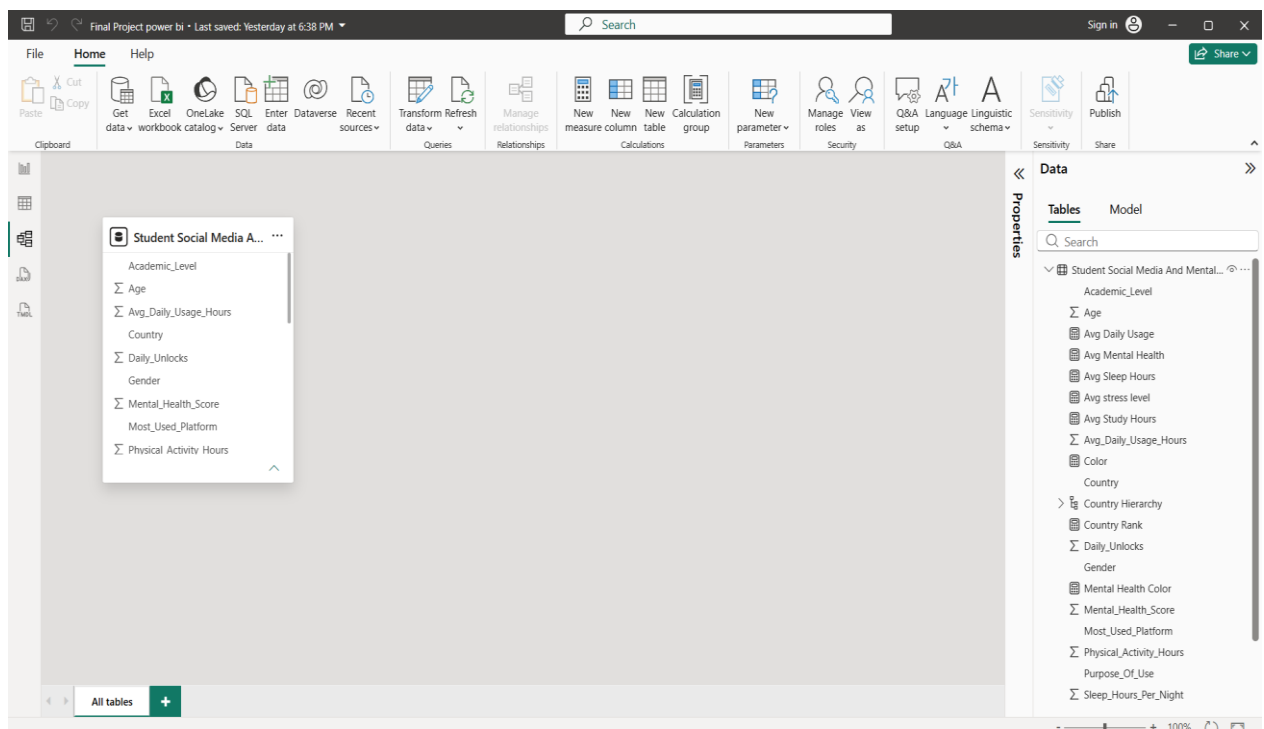
MODEL BENEFITS

Improved Performance

Easy DAX Development

Better Scalability

SCREENSHOT



DAX MEASURES USED

TOTAL STUDENTS

Total Schools = COUNT('Student Social Media And Mental Health Impact'[Academic_Level])

PURPOSE : Calculate total Students

AVG DAILY USAGE OF SOCIAL MEDIA

Avg Daily Usage = AVERAGE('Student Social Media And Mental Health Impact'[Avg_Daily_Usage_Hours])

PURPOSE : Calculate total Avg daily usage of social media

TOTAL AVG MENTAL HEALTH

Avg Mental Health = AVERAGE('Student Social Media And Mental Health Impact'[Mental_Health_Score])

PURPOSE : Calculate avg mental health of students

TOTAL AVERAGE SLEEP HOURS

Avg Sleep Hours = AVERAGE('Student Social Media And Mental Health Impact'[Sleep_Hours_Per_Night])

PURPOSE : Calculate total Sleep hours

TOTAL AVERAGE STRESS LEVEL

Avg stress level = AVERAGE('Student Social Media And Mental Health Impact'[Stress level 2])

PURPOSE : Calculate average stress level for students

TOTAL AVERAGE STUDY HOURS

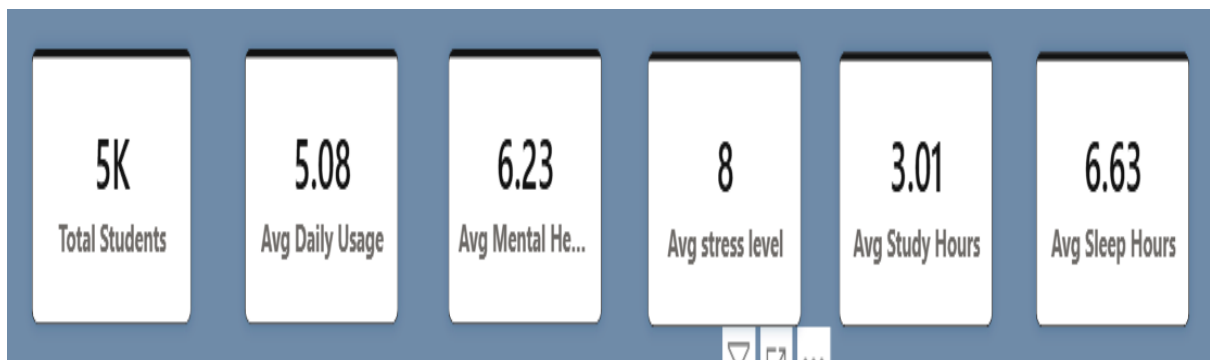
Avg Study Hours = AVERAGE('Student Social Media And Mental Health Impact'[Study_Hours])

PURPOSE : Calculate total average study hours

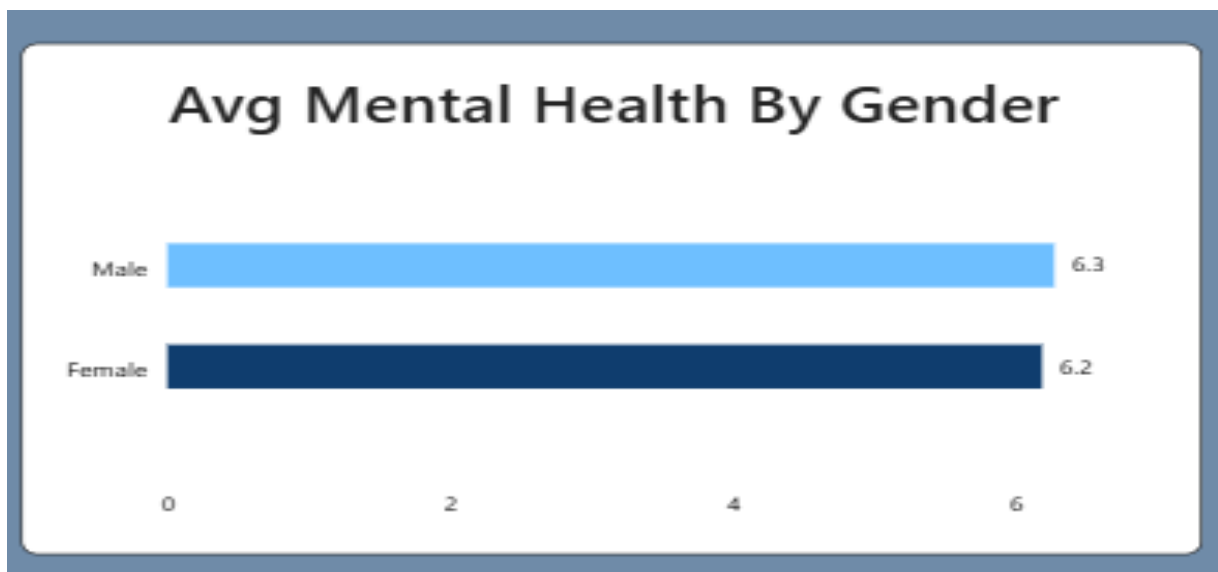
VISUALIZATIONS USED

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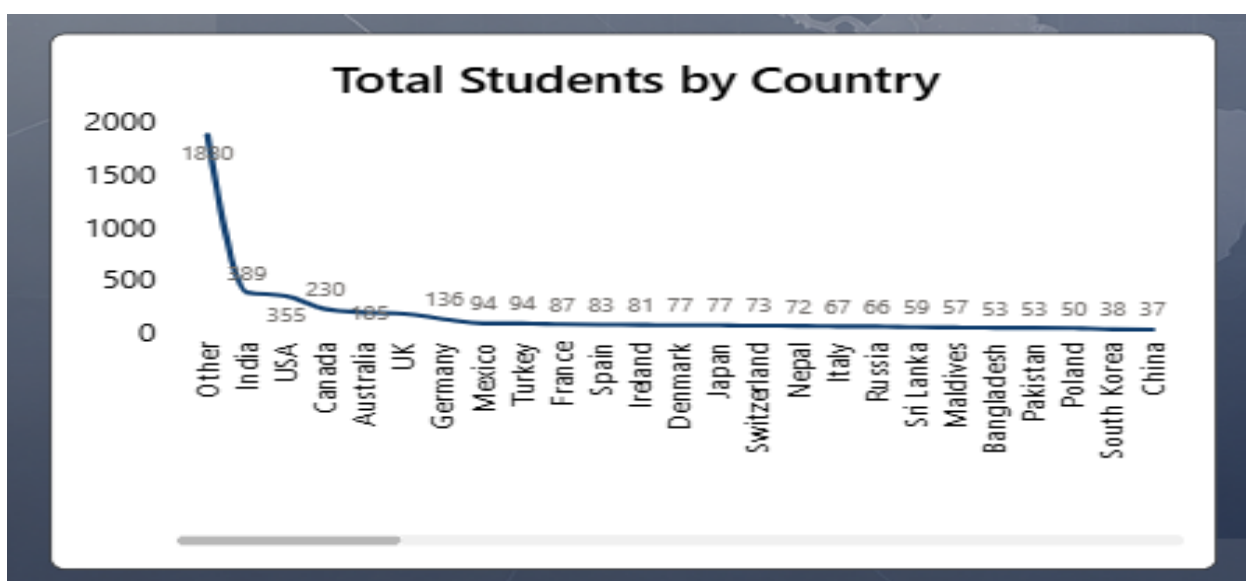
KPI CARDS:Display key mental health and social media metrics



BAR CHART: Compare avg mental health by gender



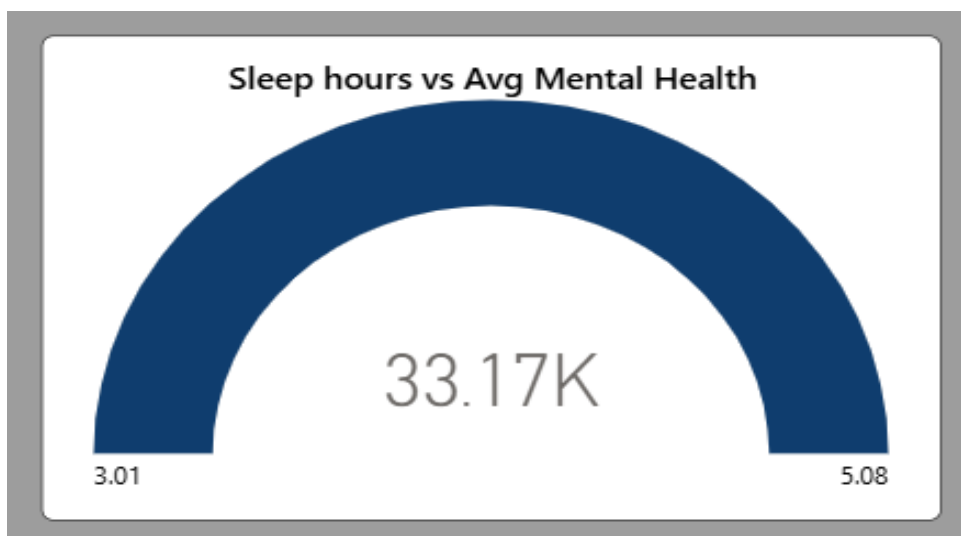
LINE CHART: Show students in country



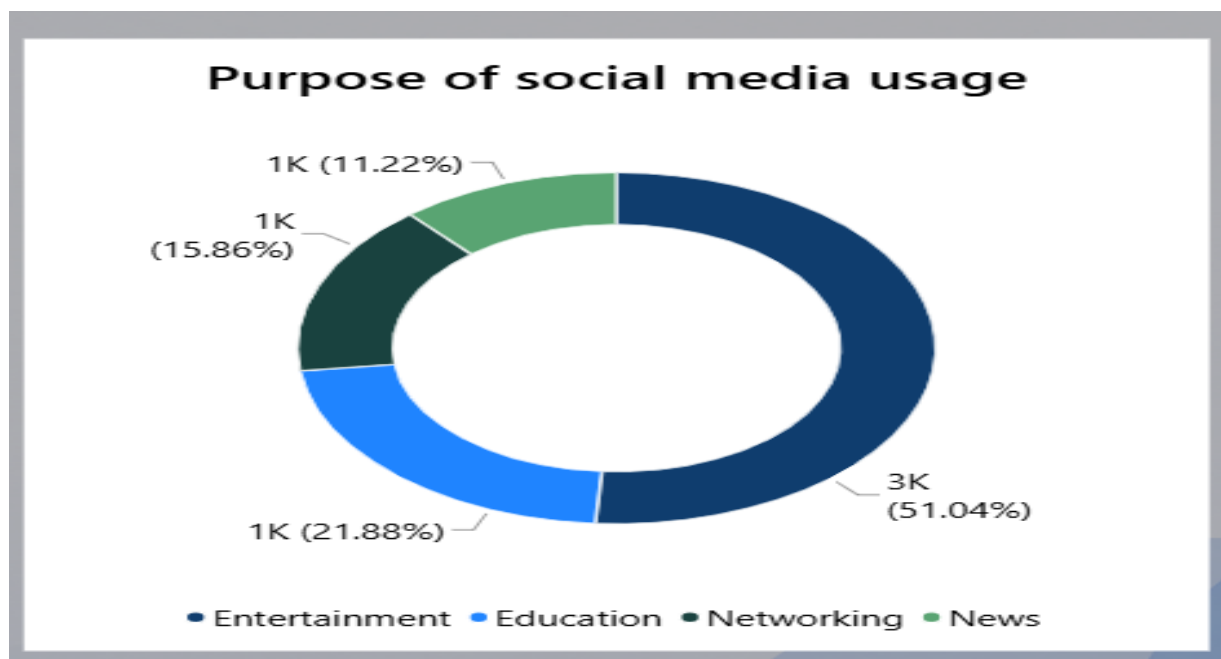
FILLEDMAP: Show Avg study by country



GAUGE CHART: Analyze Teacher-Student Ratio.

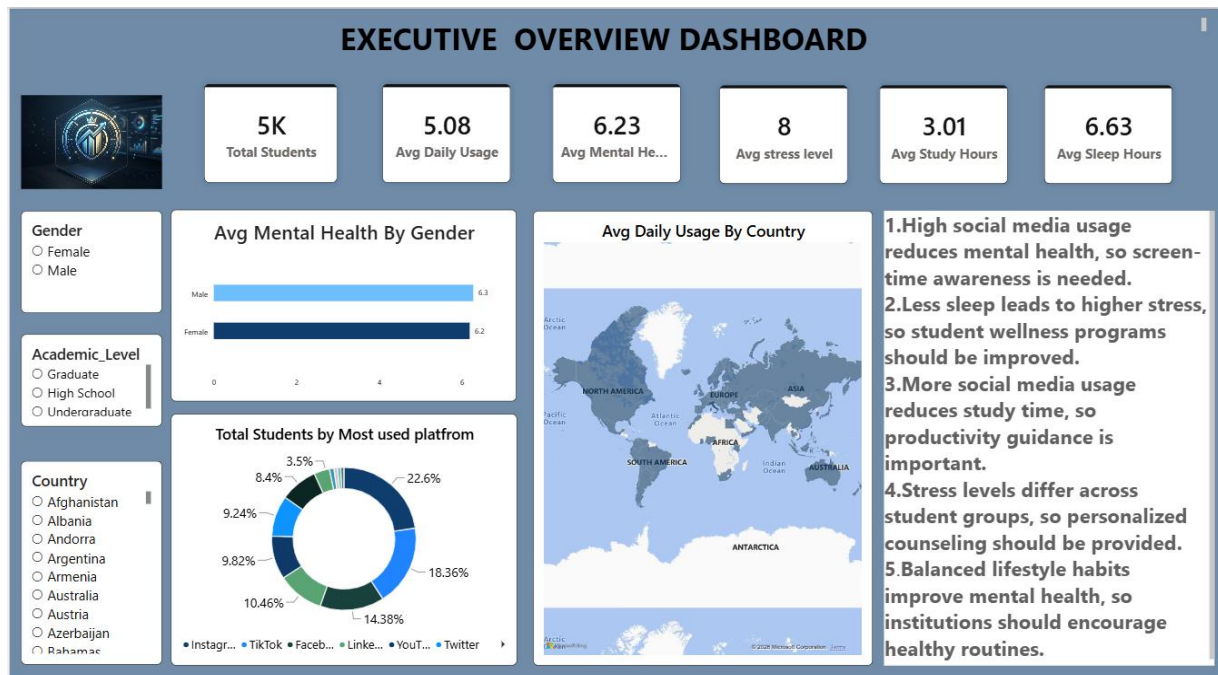


DONUT CHART: Purpose of social media usage to the total students



DASHBOARD SCREENSHOTS

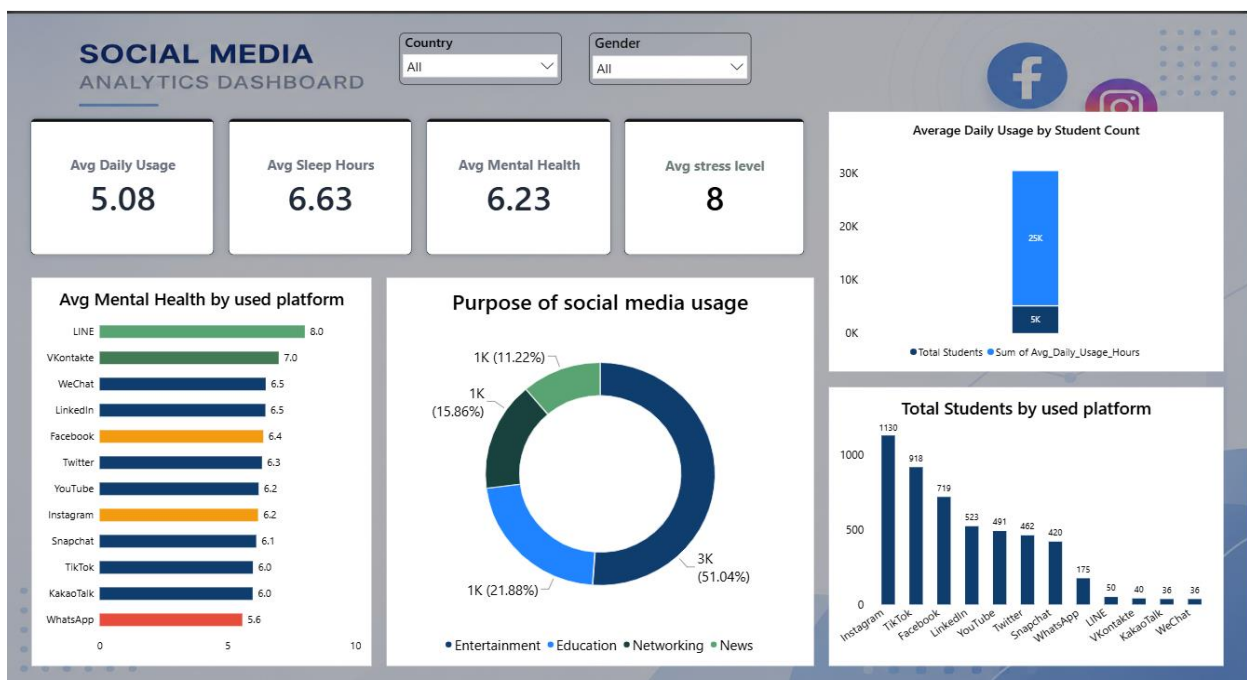
EXECUTIVE OVERVIEW



Explanation

This project analyzes how social media usage impacts students' mental health, stress, sleep, and academic performance

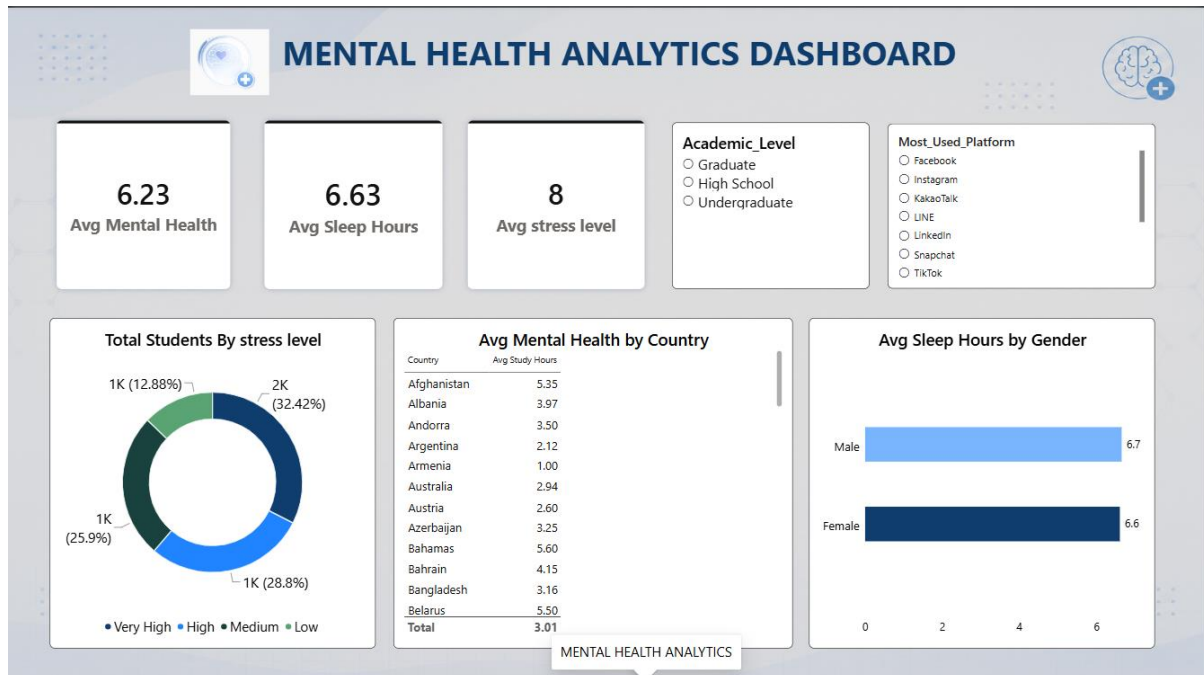
SOCIAL MEDIA ANALYTICS



Explanation

This dashboard analyzes student social media usage patterns and their impact on mental health, stress levels, and lifestyle .

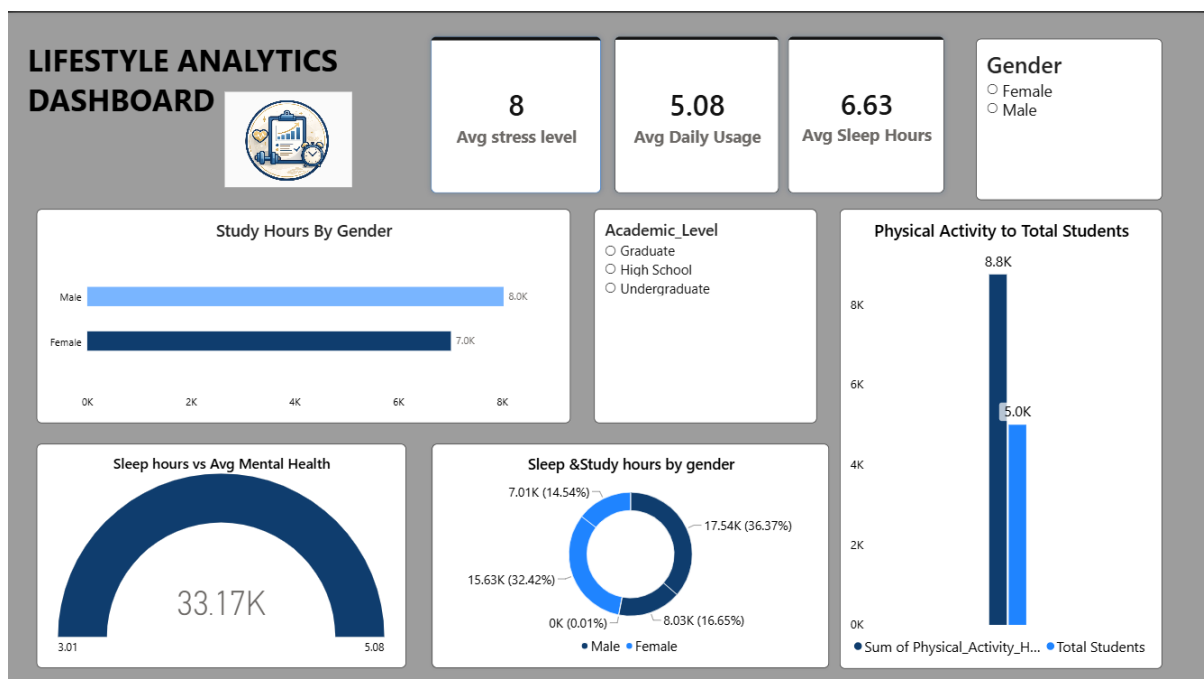
MENTAL HEALTH ANALYTICS



Explanation

This dashboard analyzes students' mental health, stress levels, sleep patterns, and country-wise well-being to identify factors affecting student wellness.

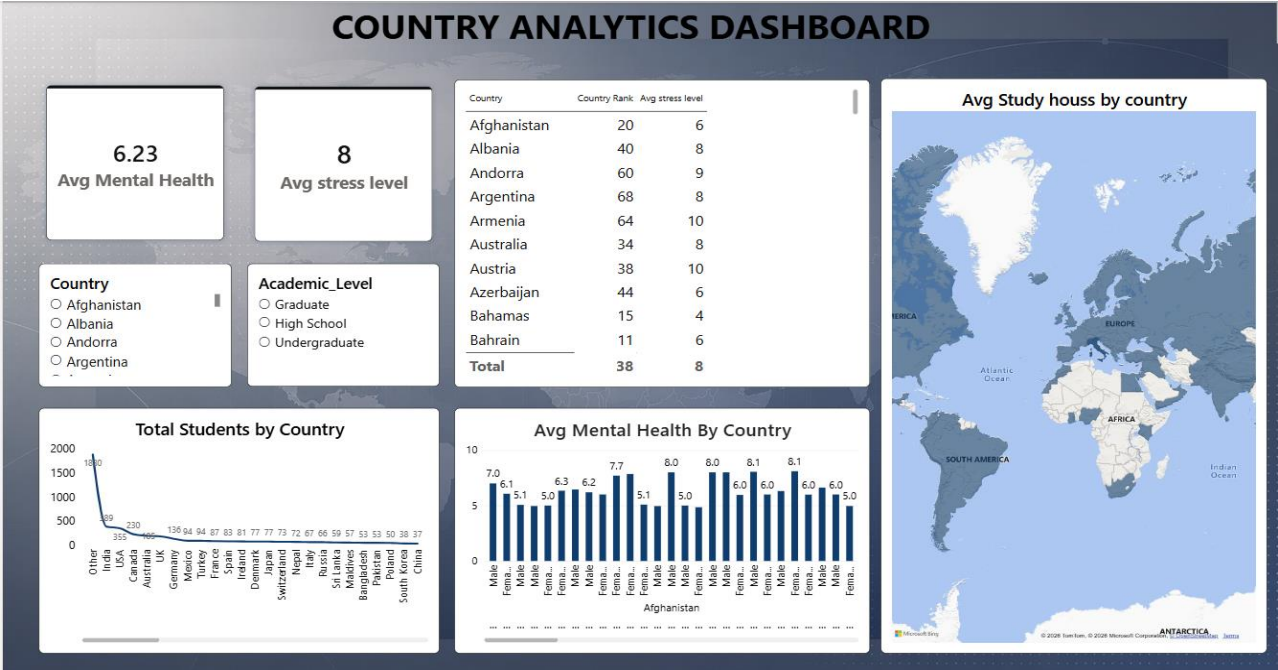
LIFE STYLE ANALYTICS



Explanation

This dashboard analyzes students’ lifestyle habits, including sleep, study time, physical activity, and stress levels to understand their impact on mental health.

COUNTRY ANALYTICS



Explanation

This dashboard analyzes student behavior across countries by comparing mental health, stress levels, study hours, and student distribution using Power BI.

KEY INSIGHTS

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INSIGHT-1

Higher social media usage negatively impacts students’ mental health and increases stress levels.

INSIGHT-2

Students with better sleep and balanced lifestyle habits show improved mental health outcomes.

INSIGHT-3

Mental health, stress levels, and student behavior vary significantly across countries and student groups.

BUSINESS RECOMMENDATIONS

RECOMMENDATION-1

Reduce excessive social media usage

RECOMMENDATION-2

Improve student sleep habits

RECOMMENDATION-3

Provide mental health counseling

RECOMMENDATION-4

Encourage healthy lifestyle routines

FEATURES IMPLEMENTED

✓ Drill Down

✓ Drill Through

✓ Tooltips

✓ Dynamic Titles

✓ Slicers

✓ Conditional Formatting

✓ Navigation Buttons

✓ Bookmarks

✓ Custom Theme

CHALLENGES FACED

Data Cleaning Challenges

Relationship Management

DAX Optimization

Dashboard Layout Design

Performance Tuning

LEARNINGS

Power Query

Data Modeling

DAX Development

Dashboard Design

Storytelling

Business Analysis

PROJECT STORYTELLING

BUSINESS SCENARIO

Educational institutions and student counselors collect large amounts of data related to students' social media usage, mental health, stress levels, sleep patterns, and study habits. However, decision-makers often struggle to transform this data into meaningful insights that support student well-being, academic performance, and mental health improvement.

BUSINESS PROBLEM

- How does social media usage affect students' mental health?
- Which social media platforms are most used by students?
- How do stress levels vary across student groups and countries?
- How do sleep patterns and study habits impact mental health?
- Which student groups require additional mental health support?

ANALYSIS APPROACH

- **Analyzed** social media usage patterns across different platforms and student groups.
- **Compared** mental health, stress levels, sleep hours, and study habits among students.
- **Evaluated** country-wise and academic-level differences in student well-being.
- **Identified** key factors affecting student mental health and lifestyle habits.

KEY FINDINGS

The analysis revealed that higher social media usage is associated with lower mental health scores and increased stress levels among students. Students with better sleep patterns and balanced study hours showed improved mental well-being and healthier lifestyles. Additionally, mental health, stress levels, and social media usage varied across different countries, academic levels, and student groups, highlighting the need for personalized wellness support and awareness programs.

BUSINESS IMPACT

- Improves student mental health awareness
- Supports better counseling decisions
- Helps identify high-risk student groups
- Encourages healthy screen-time habits
- Improves student well-being and academic performance

PROJECT DEPLOYMENT

DEPLOYMENT OBJECTIVE

Deploy the Power BI solution to enable educational stakeholders to access interactive dashboards and insights in real time.

DEPLOYMENT PLATFORM

Power BI Service

DEPLOYMENT PROCESS

Power BI Service

STEP-1

Validate Data Model and DAX Measures in Power BI Desktop.

STEP-2

Optimize Dashboard Layout, Performance, and User Experience.

STEP-3

Publish PBIX File to Power BI Service.

STEP-4

Configure Dataset Refresh Settings.

STEP-5

Create Dashboard from Published Reports.

STEP-6

Apply Security and Access Permissions.

STEP-7

Share Dashboard with Educational Stakeholders.

BUSINESS BENEFITS

Centralized Reporting Platform.

Interactive Education Analytics.

Real-Time Monitoring and Reporting.

Improved Educational Planning.

Faster Data-Driven Decisions.

CONCLUSION

This project successfully analyzed the impact of social media usage on students' mental health, stress levels, sleep patterns, and study habits using Power BI dashboards. The findings showed that excessive social media usage and poor lifestyle habits negatively affect student well-being and academic performance. The dashboard provides meaningful insights that can help educational institutions improve student wellness, mental health support, and informed decision-making.